BRAKES—BRAKE PULL—4X2 RANGER VEHICLES ONLY	Article No. 02-10-10

FORD: 2001-2002 RANGER

ISSUE

Some 4x2 Rangers may exhibit a pull to right or left while braking. Brake pull is a complex condition involving the suspension geometry, changing camber under load during braking and is extremely sensitive to road surface. It may be caused by tire mismatch, alignment, brake imbalance, damaged, worn or loose parts or possibly the lower control arm bushings used in production.

ACTION

Refer to the following Diagnostic Procedures/Tips to help diagnose and correct brake pull.

SERVICE PROCEDURE

NOTE
THE FOLLOWING STEPS SHOULD BE PERFORMED IN THE GENERAL ORDER AS LISTED.

Diagnostic Procedure 1

1. Set tire pressures to specification (listed inside door). Check for uneven tire wear.
2. Check for loose, damaged or excessively worn parts. Visually inspect for front or rear brake wear, contamination, leaks etc. Service as required. Visually inspect front suspension for loose or worn components (ie: ball joints, control arm bushings, alignment adjusters, steering linkage, etc.) Service as required.
3. Check ride height, side to side, in the front of the vehicle (ie: possible weak spring on one side). Check alignment against published specification. Loosen all four U-bolts on each side of the rear axle, make sure the axle is at a zero thrust angle, and re-torque all bolts to specification: 103 N•m (76 Lb-ft).
4. Bleed the brake system to remove any trapped air.

If no obvious concerns are found from Steps 1-3, continue with further diagnosis.

Diagnostic Procedure 2

1. Disable the ABS for diagnostic purposes (momentary/uncommanded ABS can cause a brake pull). Inspect the tone rings for damage (ie: bent, chipped, missing teeth). Check wheel speed sensor output with WDS oscilloscope or HZ meter (watch for signal drop out, indicating lock up). Replace failed components as necessary.
2. Check for front wheel rotational drag with warm/hot brakes:
   a. Make a series of 10 stops from speeds of 48 km/h (30 mph).
   b. Make the 10 stops with no more than 30 second intervals between stops. This generally will induce a maximum pull if there is a pull condition.
   c. Raise the front end of the vehicle so that both tires are off the ground.
   d. Rotate the front wheels by hand. Both tires should rotate about the same amount, with little brake drag occurring. Some brake drag will naturally be present.
3. If the vehicle has excess brake drag, proceed as follows:
   a. Open the bleed screw on the affected wheel to check for residual pressure in the brake lines. If there is a squirt of brake fluid out of the bleeder screw and the drag is reduced or eliminated, the master cylinder may not be allowing the brake fluid to return to the reservoir.
   b. Check the master cylinder push rod adjustment. Refer to section 206-07 of the appropriate model year Workshop Manual for service details.
   c. Check the brake pedal to determine if it is fully returning.

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4. Remove both calipers and check piston retraction using a C-clamp and wooden block. The piston should easily move back into the bore under the force of the clamp. Service as needed.

NOTE
LUBRICATE THE CALIPER SLIDES, MAKING SURE NOT TO GET ANY GREASE ON FRICTION MATERIAL OR THE PAD SLIPPERS. THIS CAN AGGRAVATE OR CAUSE A PULL.

5. If Steps 1 through 4 have not determined the condition, swap brake pads and rotors side-to-side and if pull changes from right to left or vice versa; machine the rotors, put a non-directional finish on the rotors, install new pads, burnish the brakes, and retest.

6. Test drive the vehicle with 10 or more stops from 64 km/h (40 mph) before evaluating brake drift. If there is no change while braking, it may be assumed the condition is not the result of the front brakes.

7. Check the rear brakes. Remove the rear wheels and brake drums.

   a. Check for contaminants (ie: grease, axle lube, leaking brake fluid, etc.)

   b. Check for broken components, frozen parking brake cables or improperly installed parts.

   c. Check the cage diameter using the Brake Shoe Adjustment Gauge (D81L-1103-A). If either side is out of specification, check for proper function of the adjusting mechanism and service as required.

   d. Check the rear wheel cylinders by the following method:

      (1) Re-install one of the brake drums.

      (2) Have another person apply the brakes and watch for movement of one of the brake shoes on the other wheel.

      (3) Reapply the brakes while holding in place the shoe that moved. The opposite shoe should now move. If no movement is observed, replace the wheel cylinder.

      (4) Reinstall the drum and repeat the procedure for the other rear brake.

8. If the condition still exists, swap rear shoes from side-to-side and drive the vehicle. If the pull is significantly reduced, replace the rear linings.

If the pull is not significantly reduced, go to Diagnostic Procedure 3.

Diagnostic Procedure 3

1. Check upper control arm shims - front and rear shims on each control arm on both left and right sides of the vehicle. Make sure the shims are seated properly and torqued to spec: 133 N•m (98 Lb-ft.).

2. Check the front stabilizer bar. The blue paint mark is on the left hand (driver's side) of the vehicle. If it is not, then the stabilizer bar was installed upside down. Reinstall correctly.

3. Replace the Lower Control Arms (LCA) and double check alignment. Use LCA (right) Part F87Z-3078-AA and LCA (left) Part F87Z-3079-AA (these LCA's have harder bushings).

PART NUMBER	PART NAME
F87Z-3078-AA	Lower Control Arm - Right
F87Z-3079-AA	Lower Control Arm - Left

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 301000, 303000, 304000